

# Shahana Immersive Store - Complete Implementation Plan

## Critical Fixes + Three Layout System + Bundle Generation

**Date:** May 29, 2026

**Status:** Ready for Hermes Agent Implementation

**Estimated Time:** 2-4 hours

**Branch:** feature/immersive-critical-fixes-and-layouts

### Executive Summary

This plan consolidates:

1. **8 Critical Issues** from immersive-core.js, immersive-features.js, immersive-init.js
2. **Three Layout System** (Asymmetric Gallery, Scroll Narrative, Masonry Featured)
3. **Liquid Template Updates** for merchant configuration
4. **Bundle Regeneration** from source files

**Files to Modify:** 5 total

- JavaScript: immersive-core.js, immersive-features.js, immersive-init.js
- Liquid: immersive-canvas.liquid, immersive-webgl-gallery-config.liquid
- Build: build-immersive-bundle.sh (new)

**No New JavaScript Files** - All code integrates into existing files.

### Part 1: JavaScript Fixes & Layout System

#### Phase 1.1: Add Core Infrastructure to immersive-core.js

**Location:** After line 152 (after CODEX\_THEME\_TO\_ROOM)

**Add:**

```
JavaScript

// _____
// LISTENER REGISTRY - Prevent memory leaks
// _____
```

```

var ListenerRegistry = {
  registry: {},

  add: function(key, element, event, handler, options) {
    if (!this.registry[key]) this.registry[key] = [];
    element.addEventListener(event, handler, options);
    this.registry[key].push({ element: element, event: event, handler: handler,
  },

  cleanup: function(key) {
    if (!this.registry[key]) return;
    this.registry[key].forEach(function(listener) {
      try {
        listener.element.removeEventListener(listener.event, listener.handler,
      } catch (e) {
        if (window.__IMMERSIVE_DEV__) console.warn('[Immersive] Listener removal failed');
      }
    });
    delete this.registry[key];
  },

  cleanupAll: function() {
    var self = this;
    Object.keys(this.registry).forEach(function(key) {
      self.cleanup(key);
    });
  }
};

// -----
// IMMERSIVE STORE NAMESPACE - Prevent global pollution
// -----

if (!window.ShahanaImmersive) {
  window.ShahanaImmersive = {
    graphics: {
      renderer: null,
      scene: null,
      camera: null,
      planeMesh: null,
      uniforms: null
    },
    room: {
      current: null,
      subMode: null,
      transitioning: false
    },
    cache: {
      textures: [],

```

```

        content: {},
        gallery: {}
    },
    settings: {
        reduceMotion: false,
        isMobile: false,
        isTablet: false,
        textureQuality: 1.0,
        targetFPS: 60
    },
    search: {
        activeIndex: -1,
        results: [],
        debounceTimer: null,
        abortController: null
    },
    gesture: {
        lastRoomTransition: 0,
        cooldown: 600,
        roomSequence: ['storefront', 'lounge', 'designer_houses', 'occasions', 'f
    },
    quickAdd: {
        modal: null,
        trigger: null
    },
    hotspot: {
        elements: [],
        focusedIndex: -1
    },
    device: {
        isMobile: false,
        isTablet: false,
        isLowEnd: false
    },
    layout: {
        registry: {},
        current: {}
    }
};
}

```

```

// -----
// DEVICE OPTIMIZATION
// -----
function initDeviceOptimization() {
    var isMobile = /iPhone|iPad|Android|Mobile/.test(navigator.userAgent);
    var isTablet = /iPad|Android/.test(navigator.userAgent) && !/Mobile/.test(nav
    var cores = navigator.hardwareConcurrency || 1;

```

```

var memory = navigator.deviceMemory || 4;
var isLowEnd = cores <= 2 || memory <= 2;

window.ShahanaImmersive.device.isMobile = isMobile;
window.ShahanaImmersive.device.isTablet = isTablet;
window.ShahanaImmersive.device.isLowEnd = isLowEnd;
window.ShahanaImmersive.settings.isMobile = isMobile;
window.ShahanaImmersive.settings.isTablet = isTablet;

if (isLowEnd) {
    window.ShahanaImmersive.settings.textureQuality = 0.5;
    window.ShahanaImmersive.settings.targetFPS = 24;
} else if (isMobile) {
    window.ShahanaImmersive.settings.textureQuality = 0.75;
    window.ShahanaImmersive.settings.targetFPS = 30;
} else {
    window.ShahanaImmersive.settings.textureQuality = 1.0;
    window.ShahanaImmersive.settings.targetFPS = 60;
}

if (window.__IMMERSIVE_DEV__) {
    console.log('[Immersive] Device optimization initialized:', {
        isMobile: isMobile,
        isTablet: isTablet,
        isLowEnd: isLowEnd,
        textureQuality: window.ShahanaImmersive.settings.textureQuality,
        targetFPS: window.ShahanaImmersive.settings.targetFPS
    });
}
}

// -----
// LOADING STATE MANAGEMENT
// -----
var LoadingState = {
    isLoading: false,
    startTime: null,
    timeout: 15000,
    timeoutId: null,

    start: function() {
        this.isLoading = true;
        this.startTime = Date.now();
        var self = this;
        this.timeoutId = setTimeout(function() {
            if (self.isLoading) {
                self.fail('Loading took too long. Please check your connection.');
            }
        }

```

```

    }, this.timeout);
  },

  complete: function() {
    this.isLoading = false;
    clearTimeout(this.timeoutId);
  },

  fail: function(message) {
    this.isLoading = false;
    clearTimeout(this.timeoutId);
    if (typeof showFeedback === 'function') {
      showFeedback(message, 'error');
    }
  }
};

// -----
// ANALYTICS
// -----
var Analytics = {
  track: function(event, data) {
    try {
      if (window.gtag) {
        gtag('event', event, data);
      }
      if (window.shopifyAnalytics) {
        shopifyAnalytics.publish('immersive:' + event, data);
      }
    } catch (e) {
      if (window.__IMMERSIVE_DEV__) console.warn('[Immersive] Analytics error: '
    }
  },

  trackRoomTransition: function(fromRoom, toRoom) {
    this.track('room_transition', { from: fromRoom, to: toRoom, timestamp: Date.now() });
  },

  trackError: function(errorType, message) {
    this.track('error', { type: errorType, message: message, timestamp: Date.now() });
  }
};

// -----
// LAYOUT SYSTEM - Three Configurable Layouts
// -----
var LAYOUT_REGISTRY = {
  'asymmetric-gallery': {

```

```

    name: 'Asymmetric Gallery',
    description: 'Indrajaal-inspired organic gallery layout for Designer Houses',
    rooms: ['designer_houses'],
    config: {
      spacing: 3.5,
      maxItems: 12,
      draggable: true,
      physics: true,
      parallaxStrength: 0.08
    }
  },
  'scroll-narrative': {
    name: 'Scroll Narrative',
    description: 'Story-driven vertical scroll layout for Occasions',
    rooms: ['occasions'],
    config: {
      spacing: 2.5,
      maxItems: 8,
      draggable: false,
      physics: false,
      parallaxStrength: 0.04,
      scrollDriven: true
    }
  },
  'masonry-featured': {
    name: 'Masonry Featured',
    description: 'Curated grid with featured highlight for Collections',
    rooms: ['featured_collections'],
    config: {
      spacing: 3.0,
      maxItems: 9,
      draggable: true,
      physics: false,
      parallaxStrength: 0.06,
      featuredIndex: 0
    }
  }
};

function getLayoutForRoom(roomKey) {
  for (var layoutKey in LAYOUT_REGISTRY) {
    var layout = LAYOUT_REGISTRY[layoutKey];
    if (layout.rooms.indexOf(roomKey) !== -1) {
      return layout;
    }
  }
  return null;
}

```

```
function applyLayoutToRoom(roomKey, layoutKey) {
  if (!LAYOUT_REGISTRY[layoutKey]) {
    console.error('[Immersive] Layout not found:', layoutKey);
    return false;
  }
  window.ShahanaImmersive.layout.current[roomKey] = layoutKey;
  if (window.__IMMERSIVE_DEV__) {
    console.log('[Immersive] Applied layout', layoutKey, 'to room', roomKey);
  }
  return true;
}
```

## Phase 1.2: Update Gallery Stage Disposal

**Location:** In `disposeGalleryStage()` function (around line 1000+)

**Replace:** The entire texture disposal section with:

JavaScript

```
function disposeGalleryStage(roomKey) {
  var state = galleryStageRegistry[roomKey];
  if (!state) return;

  // Track cleanup for analytics
  Analytics.track('gallery_stage_disposed', { room: roomKey, timestamp: Date.now() });

  if (state.group) {
    state.group.traverse(function (obj) {
      if (obj.isMesh) {
        if (obj.geometry) {
          try { obj.geometry.dispose(); } catch (e) {}
        }
        if (obj.material) {
          if (Array.isArray(obj.material)) {
            obj.material.forEach(function (m) {
              if (m.map) { try { m.map.dispose(); } catch (e) {} }
              try { m.dispose(); } catch (e) {}
            });
          } else {
            if (obj.material.map) { try { obj.material.map.dispose(); } catch (e) {} }
            try { obj.material.dispose(); } catch (e) {}
          }
        }
      }
    });
  }
  try { scene.remove(state.group); } catch (e) {}
}
```

```

}

// Dispose all textures
if (state.textures) {
  state.textures.forEach(function (tex) {
    if (tex) {
      try {
        tex.dispose();
      } catch (e) {
        if (window.__IMMERSIVE_DEV__) console.warn('[Immersive] Texture dispo
      }
    }
  });
  state.textures = [];
}

// Clean up listeners
ListenerRegistry.cleanup('gallery-' + roomKey);

// Clear state
state.dragState = null;
state.scrollState = null;
state.interactionListeners = null;
state.group = null;

delete galleryStageRegistry[roomKey];
}

```

## Phase 1.3: Initialize Device Optimization

**Location:** After line 900 (after global variable declarations)

**Add:**

JavaScript

```

// Initialize device optimization on load
initDeviceOptimization();

```

## Phase 1.4: Update immersive-init.js - Fix Event Listener Leaks

**Critical Issues to Fix:**

1. **FAB Event Listener Leak** (line ~1021-1249)
  - Replace all `addEventListener` with `ListenerRegistry.add()`
2. **Quick Add Modal Leak** (line ~603-791)

- Wrap listener cleanup in modal close handler

### 3. Search Debounce (line ~177-500)

- Use AbortController for fetch requests
- Cancel debounce timer on search close

### 4. Gesture Listeners (line ~450-543)

- Use ListenerRegistry for all touch/gesture listeners

#### Template for fixes:

JavaScript

```
// OLD (LEAK):
element.addEventListener('click', handler);

// NEW (FIXED):
ListenerRegistry.add('fab-trigger', element, 'click', handler);

// On cleanup:
ListenerRegistry.cleanup('fab-trigger');
```

## Phase 1.5: Update immersive-features.js - Error Handling

Add try-catch blocks around:

1. Texture loading (line ~500+)
2. Product panel opening (line ~600+)
3. Collection panel opening (line ~700+)

#### Template:

JavaScript

```
try {
  // existing code
} catch (e) {
  Analytics.trackError('feature_error', e.message);
  if (window.__IMMERSIVE_DEV__) console.error('[Immersive]', e);
  LoadingState.fail('An error occurred. Please try again.');
```

## Part 2: Liquid Template Updates

## Phase 2.1: Update immersive-canvas.liquid Schema

**Location:** Line 2681-2689 (gallery\_layout setting)

**Replace:**

Plain Text

```
{
  "type": "select",
  "id": "gallery_layout",
  "label": "Gallery Layout",
  "info": "Layout for the 3D gallery cards shown in rooms with collection slots",
  "default": "vertical",
  "options": [
    { "value": "vertical", "label": "Vertical Scroll" },
    { "value": "arc", "label": "Arc Carousel" }
  ]
}
```

**With:**

Plain Text

```
{
  "type": "header",
  "content": "Gallery Layouts"
},
{
  "type": "select",
  "id": "designer_houses_layout",
  "label": "Designer Houses Layout",
  "default": "asymmetric-gallery",
  "options": [
    { "value": "asymmetric-gallery", "label": "Asymmetric Gallery (Indrajaal-st" },
    { "value": "scroll-narrative", "label": "Scroll Narrative" },
    { "value": "masonry-featured", "label": "Masonry Featured" }
  ]
},
{
  "type": "select",
  "id": "occasions_layout",
  "label": "Occasions Layout",
  "default": "scroll-narrative",
  "options": [
    { "value": "asymmetric-gallery", "label": "Asymmetric Gallery" },
    { "value": "scroll-narrative", "label": "Scroll Narrative (Story-driven)" },
    { "value": "masonry-featured", "label": "Masonry Featured" }
  ]
}
```

```

]
},
{
  "type": "select",
  "id": "featured_collections_layout",
  "label": "Featured Collections Layout",
  "default": "masonry-featured",
  "options": [
    { "value": "asymmetric-gallery", "label": "Asymmetric Gallery" },
    { "value": "scroll-narrative", "label": "Scroll Narrative" },
    { "value": "masonry-featured", "label": "Masonry Featured (Curated)" }
  ]
}

```

## Phase 2.2: Pass Layout Settings to JavaScript

**Location:** Line 32 in immersive-canvas.liquid

**Replace:**

Plain Text

```
data-gallery-layout="{{ section.settings.gallery_layout | default: 'vertical' }}"
```

**With:**

Plain Text

```

data-designer-houses-layout="{{ section.settings.designer_houses_layout | default: 'vertical' }}"
data-occasions-layout="{{ section.settings.occasions_layout | default: 'scroll-narrative' }}"
data-featured-collections-layout="{{ section.settings.featured_collections_layout | default: 'masonry-featured' }}"

```

## Phase 2.3: Update immersive-webgl-gallery-config.liquid

**Location:** Line 21 (data-room-key)

**Add layout attribute:**

Plain Text

```

data-room-key="{{ section.settings.room_key }}"
data-layout="{{ section.settings.layout | default: 'default' }}"

```

**Add to schema** (after room\_key setting):

Plain Text

```
{
  "type": "select",
  "id": "layout",
  "label": "Gallery Layout for this room",
  "options": [
    { "value": "default", "label": "Default" },
    { "value": "asymmetric-gallery", "label": "Asymmetric Gallery" },
    { "value": "scroll-narrative", "label": "Scroll Narrative" },
    { "value": "masonry-featured", "label": "Masonry Featured" }
  ],
  "default": "default"
}
```

---

## Part 3: Build Process

### Phase 3.1: Create Build Script

**File:** `build-immersive-bundle.sh` (already created)

**Run:**

Bash

```
cd /home/ubuntu/shahana-dawn-custom
./build-immersive-bundle.sh
```

**Output:** Regenerates `assets/immersive-bundle.js` from source files

---

## Implementation Checklist

### JavaScript Fixes

- ☐ Add ListenerRegistry to immersive-core.js
- ☐ Add ShahanaImmersive namespace
- ☐ Add device optimization
- ☐ Add LoadingState management
- ☐ Add Analytics tracking
- ☐ Add LAYOUT\_REGISTRY

- ☐ Update disposeGalleryStage() with error handling
- ☐ Fix FAB listener leaks in immersive-init.js
- ☐ Fix Quick Add modal leaks in immersive-init.js
- ☐ Fix search error handling in immersive-init.js
- ☐ Fix gesture listener leaks in immersive-init.js
- ☐ Add try-catch blocks in immersive-features.js

## Liquid Updates

- ☐ Update immersive-canvas.liquid schema (add 3 layout selectors)
- ☐ Update data attributes in immersive-canvas.liquid
- ☐ Update immersive-webgl-gallery-config.liquid schema
- ☐ Add layout attribute to gallery config

## Build & Testing

- ☐ Run build script to regenerate bundle
- ☐ Verify bundle file size (should be ~8,500-9,000 lines)
- ☐ Test on desktop (60 FPS target)
- ☐ Test on mobile (30+ FPS target)
- ☐ Test all three layouts
- ☐ Verify no memory leaks
- ☐ Verify analytics tracking

## Git & Deployment

- ☐ Commit all changes
- ☐ Create pull request
- ☐ Test in staging environment
- ☐ Deploy to production

---

## Success Criteria

- ☒ Performance

- Desktop: 60 FPS sustained
- Mobile: 30+ FPS sustained
- No memory leaks after 10+ room transitions
- Texture memory stable (<200MB)

#### ✓ **Stability**

- No crashes on room transitions
- Error recovery working
- Loading state visible
- Analytics tracking working

#### ✓ **Features**

- Three layouts rendering correctly
- Merchants can select layouts per room
- Layouts persist on page reload
- Mobile responsive

#### ✓ **Code Quality**

- All listeners properly cleaned up
- No global variable pollution
- Error handling comprehensive
- Analytics tracking comprehensive

## Risk Assessment

| Risk                    | Probability | Mitigation                           |
|-------------------------|-------------|--------------------------------------|
| Memory leak regression  | Low         | ListenerRegistry + disposal testing  |
| Performance degradation | Low         | Device optimization + FPS monitoring |
| Bundle size increase    | Low         | ~1,000 lines added, well-organized   |

|                    |     |                                    |
|--------------------|-----|------------------------------------|
| Merchant confusion | Low | Clear UI labels, defaults provided |
|--------------------|-----|------------------------------------|

---

## Notes for Hermes Agent

1. **Modify source files first**, then run build script
2. **Test each phase** before moving to next
3. **Use ListenerRegistry** for ALL event listeners
4. **Wrap with try-catch** for error resilience
5. **Track with Analytics** for debugging
6. **Test on real devices** (not just desktop)
7. **Commit frequently** with clear messages

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**Ready for Implementation** ✓